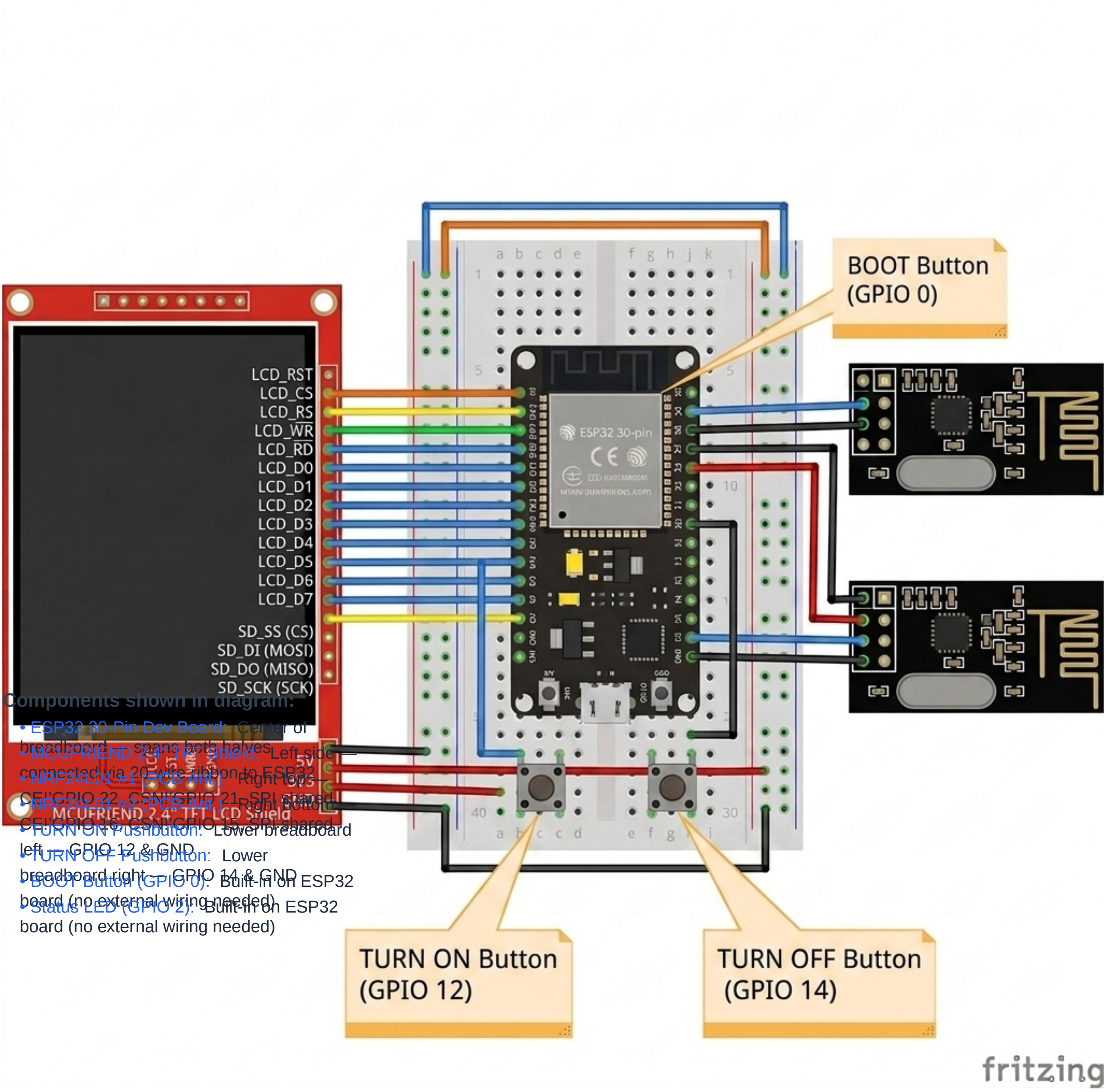
ESP32 BT JAMMER & WI-FI SECURITY SUITE

Master Hardware Pinout, RF Transceiver & Wiring Diagram Guide

LEGAL & EDUCATIONAL DISCLAIMER:

This device design and documentation are strictly for educational and authorized penetration testing in controlled environments. Operating RF jammers or packet injection without authorization is illegal.

1. Full Master Hardware Breadboard Wiring Diagram



Components shown in diagram:

- ESP32 30-Pin Dev Board: Center of breadboard — spans both halves
- MCUFRIEND 2.4" TFT Shield: Left side — connected via 20-wire ribbon to ESP32
- NRF24L01 #1 (PCB Ant.): Right top — CE!’GPIO 22, CSN!’GPIO 21, SPI shared
- NRF24L01 #2 (PCB Ant.): Right bottom — CE!’GPIO 16, CSN!’GPIO 15, SPI shared
- TURN ON Pushbutton: Lower breadboard left — GPIO 12 & GND
- TURN OFF Pushbutton: Lower breadboard right — GPIO 14 & GND
- BOOT Button (GPIO 0): Built-in on ESP32 board (no external wiring needed)
- Status LED (GPIO 2): Built-in on ESP32 board (no external wiring needed)

1B. Breadboard Wire-by-Wire Connection Reference

Wire Color Key:

- **RED** 3.3V / 5V Power Supply Rail
- **BLACK** GND Ground Rail
- **BLUE** SPI SCK (Clock) — GPIO 18
- **PURPLE** SPI MOSI (Data Out) — GPIO 23
- **CYAN** SPI MISO (Data In) — GPIO 19
- **ORANGE** NRF CE / CSN control lines
- **GREEN** TURN ON Button wire — GPIO 12
- **YELLOW** TFT control lines (RS, CS, RST, WR)
- **VIOLET** TURN OFF Button wire — GPIO 14

Complete Breadboard Connection Table — All Wires

From Component / Pin	ESP32 GPIO	Wire Color	Destination / Rail
Breadboard +Rail (3.3V)	**3.3V pin**	RED	!' NRF24 #1 VCC, NRF24 #2 VCC
Breadboard " Rail (GND)	**GND pin**	BLACK	!' NRF24 #1 GND, NRF24 #2 GND, BTN GND legs
TFT Shield 5V/3V3	**5V / 3.3V**	RED	!' Breadboard power rail
TFT Shield GND	**GND**	BLACK	!' Breadboard GND rail
SPI Clock (SCK)	**GPIO 18**	BLUE	!' NRF24 #1 SCK, NRF24 #2 SCK, TFT SD_SCK
SPI MOSI	**GPIO 23**	PURPLE	!' NRF24 #1 MOSI, NRF24 #2 MOSI, TFT SD_DI
SPI MISO	**GPIO 19**	CYAN	!' NRF24 #1 MISO, NRF24 #2 MISO, TFT SD_DO
NRF24 #1 CE	**GPIO 22**	ORANGE	!' NRF24 #1 CE pin
NRF24 #1 CSN	**GPIO 21**	ORANGE	!' NRF24 #1 CSN pin
NRF24 #2 CE	**GPIO 16**	ORANGE	!' NRF24 #2 CE pin
NRF24 #2 CSN	**GPIO 15**	ORANGE	!' NRF24 #2 CSN pin
TFT LCD_RST	**GPIO 33**	YELLOW	!' TFT Shield LCD_RST pin
TFT LCD_CS	**GPIO 5**	YELLOW	!' TFT Shield LCD_CS pin
TFT LCD_RS (DC)	**GPIO 4**	YELLOW	!' TFT Shield LCD_RS pin
TFT LCD_WR	**GPIO 2**	YELLOW	!' TFT Shield LCD_WR pin
TFT LCD_RD	**3.3V Rail**	RED	!' Tie HIGH (no data read needed)
TFT LCD_D0	**GPIO 12**	BLUE	!' TFT Shield D0 (shared w/ TURN ON btn)
TFT LCD_D1	**GPIO 13**	BLUE	!' TFT Shield D1 (shared w/ SD_SS)
TFT LCD_D2	**GPIO 26**	BLUE	!' TFT Shield D2
TFT LCD_D3	**GPIO 25**	BLUE	!' TFT Shield D3
TFT LCD_D4	**GPIO 17**	BLUE	!' TFT Shield D4
TFT LCD_D5	**GPIO 27**	BLUE	!' TFT Shield D5
TFT LCD_D6	**GPIO 14**	BLUE	!' TFT Shield D6 (shared w/ TURN OFF btn)
TFT LCD_D7	**GPIO 32**	BLUE	!' TFT Shield D7
SD Card SS (CS)	**GPIO 13**	YELLOW	!' TFT Shield SD_SS
TURN ON Pushbutton	**GPIO 12**	GREEN	Leg 1 !' GPIO 12, Leg 2 !' GND Rail
TURN OFF Pushbutton	**GPIO 14**	VIOLET	Leg 1 !' GPIO 14, Leg 2 !' GND Rail
BOOT Button	**GPIO 0**	—	Built-in on ESP32 board (no wire needed)
Status LED	**GPIO 2**	—	Built-in on ESP32 board (no wire needed)

2. ESP32 Pin Allocation Overview

ESP32 30-PIN DEV BOARD LAYOUT

Radio 1 CE	**GPIO 22**	**GPIO 21**	Radio 1 CSN
Radio 2 CE	**GPIO 16**	**GPIO 15**	Radio 2 CSN
Touch T_CS	**GPIO 27**	**GPIO 4**	TFT DC / RS
TFT Backlight	**GPIO 32**	**GPIO 5**	TFT CS
TFT Reset	**GPIO 33**	**GPIO 18**	SPI SCK (Clock)
Physical ON Btn	**GPIO 12**	**GPIO 19**	SPI MISO
Physical OFF Btn	**GPIO 14**	**GPIO 23**	SPI MOSI
Power Rail	**3.3V**	**GND**	Ground Rail

3. Hardware Interfaces (VSPI & ESP32 Wi-Fi Radio)

Signal / Interface	ESP32 Pin	Connected Modules / Mode	Type
SCK (Clock)	**GPIO 18**	Radio 1, Radio 2, TFT SCK, Touch T_CLK	Shared SPI
MISO (Master In)	**GPIO 19**	Radio 1, Radio 2, TFT SDO, Touch T_DO	Shared SPI
MOSI (Master Out)	**GPIO 23**	Radio 1, Radio 2, TFT SDI, Touch T_DIN	Shared SPI
Wi-Fi 802.11 b/g/n	**Internal RF**	AP Scanning, Promiscuous Sniffer, Web Portal	Internal RF

4. Radio Transceiver Modules (NRF24L01)

Radio #1 (Lower Channels 2402-2440 MHz):

NRF24 #1 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	**3.3V / GND**	Power Rail (+3.3V Max)	Power / Ground
CE	**GPIO 22**	Chip Enable	Dedicated Control
CSN	**GPIO 21**	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	**GPIO 18 / 23 / 19**	Hardware SPI Bus	Shared SPI

Radio #2 (Upper Channels 2441-2480 MHz):

NRF24 #2 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	**3.3V / GND**	Power Rail (+3.3V Max)	Power / Ground
CE	**GPIO 16**	Chip Enable	Dedicated Control
CSN	**GPIO 15**	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	**GPIO 18 / 23 / 19**	Hardware SPI Bus	Shared SPI

5. MCUFRIEND 2.4" TFT LCD Shield (www.mcufriend.com)

Shield Pin	ESP32 Pin	Function	Signal Category
5V / 3V3 / GND	**5V / 3.3V / GND**	Display Power & Ground	Power / Ground
LCD_RST	**GPIO 33**	Hardware Reset	Dedicated Control
LCD_CS	**GPIO 5**	Chip Select	Dedicated Control
LCD_RS	**GPIO 4**	Register / Command Select	Dedicated Control
LCD_WR	**GPIO 2**	Write Enable	Dedicated Control
LCD_RD	**3.3V (Tie HIGH)**	Read Enable (Not Used)	Power
LCD_D0	**GPIO 12**	Data Bit 0	8-Bit Parallel
LCD_D1	**GPIO 13**	Data Bit 1	8-Bit Parallel
LCD_D2	**GPIO 26**	Data Bit 2	8-Bit Parallel
LCD_D3	**GPIO 25**	Data Bit 3	8-Bit Parallel
LCD_D4	**GPIO 17**	Data Bit 4	8-Bit Parallel
LCD_D5	**GPIO 27**	Data Bit 5	8-Bit Parallel
LCD_D6	**GPIO 14**	Data Bit 6	8-Bit Parallel
LCD_D7	**GPIO 32**	Data Bit 7	8-Bit Parallel
SD_SS (CS)	**GPIO 13**	SD Card Chip Select	SPI CS
SD_DI (MOSI)	**GPIO 23**	SD SPI Data In	Shared SPI
SD_DO (MISO)	**GPIO 19**	SD SPI Data Out	Shared SPI
SD_SCK (SCK)	**GPIO 18**	SD SPI Clock	Shared SPI

6. Hardware Pushbuttons & System Controls

Button / LED	ESP32 Pin	Wiring Connection	Mode
TURN ON Button	**GPIO 12**	One leg to GPIO 12, other leg to GND	INPUT_PULLUP
TURN OFF Button	**GPIO 14**	One leg to GPIO 14, other leg to GND	INPUT_PULLUP
BOOT Button	**GPIO 0**	Built-in BOOT pushbutton on ESP32 board	Internal
Status LED	**GPIO 2**	Built-in LED on ESP32 board	OUTPUT

7. Power Supply & Capacitor Recommendations

1. Built-in PCB Antenna Modules: Draw ~12mA each. NO CAPACITORS REQUIRED! Power directly from ESP32 3.3V pin.
2. High-Power PA+LNA Modules: Draw ~115mA peak each. Requires 10uF-47uF capacitors across VCC & GND to buffer current spikes.